

Backward Uniqueness for 3D Navier–Stokes Equations With Non-Trivial Final Data and Applications

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Presented is a backward uniqueness result of bounded mild solutions of 3D Navier–Stokes Equations in the whole space with non-trivial final data. A direct consequence is that a solution must be axisymmetric in $[0, T]$ if it is so at time T . The proof is based on a new weighted estimate that enables to treat terms involving Calderon–Zygmund operators. The new weighted estimate is expected to have certain applications in control theory when classical Carleman-type inequality is not applicable.

1 Introduction

In this paper, we consider the backward uniqueness problem for the following 3D Navier–Stokes equations in $\mathbb{R}^3 \times [0, T]$:

$$\begin{cases} \partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = f, \\ \nabla \cdot u = 0, \\ u|_{t=T} = g(x), \end{cases} \quad (1.1)$$

where u denotes the velocity field, p is the pressure, and f is the forcing term. Here, we are interested in bounded mild solutions. Hence, it is assumed that the final data g is a bounded function, and the existence of the bounded mild solution u is a known fact. More precisely, one has

$$p = \mathcal{K} : (u \otimes u - F),$$

and

$$u(t) = e^{(t-s)\Delta} u(s) + \int_s^t e^{(t-\tau)\Delta} \mathbb{P}[\nabla \cdot (u \otimes u)(\cdot, \tau) + f(\cdot, \tau)] d\tau, \quad 0 \leq s \leq t \leq T,$$

with $f = \nabla \cdot F$, where $\mathcal{K} = (-\Delta)^{-1} \nabla^2$ is the standard Riesz operator and $\mathbb{P}$ is the Leray projector. For more details, see Section 2.

The backward uniqueness (BU) problem asks whether there are two different bounded mild solutions u_1 and u_2 to (1.1) with the same final data g at some time $T \in (0, +\infty)$. In this article we are interested in the case of non-trivial final data g . We say that the (BU) property holds if u_1 and u_2 must coincide.

1.1 Background

The (BU) property is closely connected to internal control theory. In fact, if u_1, u_2 are two different bounded mild solutions to (1.1), denote $u := u_2 - u_1$. It is clear that u satisfies the following modified Navier–Stokes equation with internal control:

$$\begin{cases} \partial_t u + u \cdot \nabla u - \Delta u + \nabla \pi = -u_1 \cdot \nabla u - u \cdot \nabla u_1, \\ \nabla \cdot u = 0, \\ u|_{t=t_0} = u_0. \end{cases} \quad (1.2)$$

Indeed, by forgetting about the relation between u and u_1 , one can think of u_1 as an internal control function. That the (BU) property holds means that under suitable constraints on the control functions u_1 , the null controllability for (1.2) is impossible, that is, for any nontrivial data $u(t_0)$, (1.2) cannot achieve the trivial state $u(\cdot, T) \equiv 0$ for any $T > t_0$.

Besides, the study of backward uniqueness has its own significance in the PDE's field, especially in the regularity theory of parabolic equations. For instance, the regularity theory of Navier–Stokes equations in critical spaces ([9], [13], [24], [20], [21]), regularity of harmonic maps [25], and blow up for semi-linear heat equation [19].

In the literature, the backward uniqueness and unique continuation properties of parabolic equations have been widely considered. Ladyzhenskaya [15] proved the backward uniqueness of the Navier–Stokes equations in bounded domains provided the solutions are sufficiently smooth. The proof in Ladyzhenskaya's paper is based essentially on log-convexity arguments for suitable quantities and can be applied to linear equations $u_t = A(t)u$ in Banach spaces, under suitable assumptions, see section 2 in [15] and the reference therein. Lin [18] proved the strong unique continuation for linear parabolic equations of the form $\partial_t u - \Delta u + V(x)u = 0$, where time-independent coefficients $V(x) \in L_x^{(n+1)/2}$. See also [5], [6] for results on time-dependent potentials. For equations of the form $\partial_t u - \Delta u + a(t, x)u + b(t, x) \cdot \nabla u = 0$, Poon [22] proved that if a, b are bounded, the bounded solution u in $\mathbb{R}^n \times \mathbb{R}^+$ is identically zero if it vanishes of infinite order at any point. See also [4] for improved results allowing a suitable growth condition on u and [1], [7] for analogous results on divergence-form backward operator $P = \partial_t + \partial_i(a^{ij}(x, t)\partial_j)$.

In the remarkable work [9], Escauriaza, Seregin, and Sverák proved the global regularity of $L^\infty L^3$ solutions of (1.1) with null force, where backward uniqueness arguments (proved in [8]) played a crucial role. Indeed, they proved that if a function u satisfies $|\Delta u + \partial_t u| \lesssim (|u| + |\nabla u|)$ and a certain growth condition in $\mathbb{R}^n \setminus B_R \times [0, T] := Q_{R,T}$, then $u(x, 0) \equiv 0$ in $\mathbb{R}^n \setminus B_R$ implies $u(x, t) \equiv 0$ in $Q_{R,T}$. The result can be applied to prove the vanishing of the vorticity $\omega \triangleq \nabla \times u$ in $Q_{R,T}$ for large R provided ω vanishes at $t = T$. Their proof relies on some Carleman-type inequalities, which have also been widely used in control theory. For the use of Carleman estimates, please refer to [17] for proving backward uniqueness of parabolic equations in a cone and to [24], [20], [2], and [21] for proving quantitative regularity and blowup criteria for Navier–Stokes equations.

In addition to the important role of (BU) property played in the regularity theory of Navier–Stokes Equations, we point out that the (BU) property with *non-trivial final data* studied in this article has its own obvious significance. For instance, one may ask the following question:

(Q) Suppose that $f = 0$ and $u(t, x)$ is a bounded mild solution to the Navier–Stokes equations with $u(T, x)$ being an axi-symmetric vector without swirl. Must $u(t, x)$ be axi-symmetric and swirl-free?

To answer this question, one has to first answer whether u is axi-symmetric and then apply standard backward uniqueness results after various surgeries. The proof will be presented after stating the main Theorem 1.1. Let us emphasize that a crucial difference from the trivial final data case is that one has to treat terms involving Calderón–Zygmund operators. Since the classical Carleman-type inequality involves the exponential weight that does not belong to the $\mathcal{A}_p$ class for any $p \in [1, \infty]$, we need to find new weights to overcome this essential difficulty. The new weighted estimate is expected to have certain applications in control theory when classical Carleman-type inequality is not applicable.

1.2 Main result

In this paper, we will prove the backward uniqueness of forced Navier–Stokes equations (1.1) for bounded mild solutions, with nontrivial final data. Our main result is the following theorem:

Theorem 1.1. Assume $u_1(x, t), u_2(x, t)$ are two bounded mild solutions of (1.1), such that $\nabla \times u_1$ and $\nabla \times u_2$ are also bounded. If $u_1(x, T) = u_2(x, T)$ for some $T > 0$, then $(u_1, \nabla p_1)$ must coincide with $(u_2, \nabla p_2)$ at any time $0 \leq t \leq T$.

Remark 1.1. Our result only requires the vorticity to be bounded, this assumption is weaker than several classical results, such as in [22]. It is clear that the boundedness condition on vorticity can be removed if f vanishes or has a certain regularity. For instance, if $f = \nabla \cdot F$ with $F \in L_t^\infty W_x^{1,p}$ for $3 < p \leq \infty$, then one can easily derive that $u_1, u_2 \in \mathbf{C}((0, T], W^{1,\infty}(\mathbb{R}^3))$ (see [14]).

Note that the mild assumption on the solutions in Theorem 1.1 is necessary. In view of the following simple counter-example, one cannot expect that the (BU) properties hold true for general L^∞ solutions:

Example 1.2. Let $u(x, t) = (h(t), 0, 0)$, with $h(T) = 0, h'(T) = 0$, and $h(t) \neq 0$ for $t \neq T$. Let $p(t, x) = -h'(t)x_1$, then (u, p) is a classical solution of (1.1) vanishing at $t = T$, and is obviously nontrivial.

The backward uniqueness holds for a large class of solutions of (1.1) such that the Ladyzhenskaya–Prodi–Serrin regularity criteria holds. Indeed, the boundedness of u is a (non-trivial) consequence of these conditions stated below. For instance, see [3], [9], or [23] for proofs.

Corollary 1.3. Assume $u(x, t), v(x, t)$ are mild solutions of (1.1) with $f = 0$ such that $u(x, t), v(x, t) \in L^p([0, T]; L^q(\mathbb{R}^3))$ for $2/p + 3/q = 1, 3 \leq q \leq \infty$. Then, $u(x, t)$ must coincide with $v(x, t)$ at any time $0 \leq t \leq T$, provided that $u(x, T) \equiv v(x, T)$ for some $T > 0$.

Remark 1.2. The regularity assumptions on u, v and f in Corollary 1.3 can be replaced by various functional frameworks. See [23] for a comprehensive survey on the regularity criteria of (1.1).

Now let us first assume the validity of Theorem 1.1 and present the answer of (Q).

Proof. First of all, we show that $u(t, x)$ is also axi-symmetric for all $t \in [0, T]$. Recall that a vector u is called axi-symmetric if

$$u(x) = u^r e_r + u^\theta e_\theta + u^z e_z,$$

where u^r, u^θ, u^z are all functions of (r, z) but do not depend on θ , with $r = \sqrt{x_1^2 + x_2^2}, \theta = \arctan \frac{x_2}{x_1}$, and $z = x_3$. Then any rotation by ϕ degrees around the z -axis can be described by the matrix

$$S_\phi = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is easy to check that the Navier–Stokes equation is invariant under axi-symmetric transformation. Hence if $u(x, t)$ is a solution of the N-S equation, then so does $u_\phi := S_\phi u(S_{-\phi} x, t)$. Since $u(x, T)$ is an axi-symmetric vector, it holds that

$$S_\phi u(S_{-\phi} x, T) = u(x, T),$$

or equivalently $u_\phi(x, T) = u(x, T)$ for any $\phi \in [0, 2\pi]$. By Theorem 1.1, we have $u_\phi(x, t) = u(x, t)$ for any $t < T$, which implies that $u(x, t)$ is axi-symmetric for all $\phi \in [0, 2\pi]$ and $t < T$.

Next, denote $G = \frac{u^\theta}{r}$. It is clear that G is governed by ([11, 12])

$$G_t - \Delta G - \frac{2}{r} \partial_r G + b \cdot \nabla G + \frac{2u^r}{r} G = 0,$$

with $b = u^r e_r + u^z e_z$. Since here we are dealing with bounded mild axi-symmetric solutions, it is clear that b and $\frac{u^r}{r}$ are bounded. Similarly as in [11, 16], one may look on the above equation as a five-dimensional

one and derive that

$$|G_t - \Delta_5 G| \leq C(|G| + |\nabla G|).$$

Here

$$\Delta_5 = \Delta + \frac{2}{r} \partial_r, \quad \Delta = \partial_r^2 + \frac{1}{r} \partial_r + \partial_z^2.$$

Then one may apply classical backward uniqueness result ([22]) to derive that $G \equiv 0$.

Hence, the answer to (Q) is yes. ■

1.3 Main ingredients

Now let us explain the main difficulties of this paper and the strategies used to overcome them. (a) *Invalidity of Carleman estimates.* We emphasize that the papers mentioned above do not need to treat terms involving the Calderon–Zygmund operator and hence Carleman-type inequalities can be applied. Since exponential-type weights appearing in Carleman-type estimates do not belong to any $\mathcal{A}_p$ -class for $1 < p < \infty$, one cannot apply Carleman-type estimates to treat the terms involving Calderon–Zygmund operators. Unfortunately, for (1.1) when the final data is nontrivial, the Calderon–Zygmund operators will inevitably come in.

To overcome this difficulty, we establish a new weighted estimate (see Theorem 3.1). Compared with the Carleman estimate in [8], we use the weights with polynomial spatial decay. The idea behind this is the following: the Carleman inequality in [8] is applied to the heat equation whose kernel has exponential decay. Hence, weights with exponential decay appear naturally in [8] in view of the heat kernel. In the current setting, due to the presence of the Stokes operator, whose corresponding semi-group only enjoys an algebraic decay rate, it is natural to choose a polynomial type weight $\langle x \rangle^{-k}$.

(b) *Invalidity of $\mathcal{A}_p$ weights.* Since the solutions we deal with here are bounded mild solutions and may not have any decay when x tends to infinity, the spatial weight $\langle x \rangle^{-k}$ in our proof must be integrable over the whole space, which implies $k > 3$. On the other hand, due to the appearance of the pressure term, which produces non-local effects, we have to control terms like

$$\int_{\mathbb{R}^3} \langle x \rangle^{-k} |\mathcal{R}f|^2 dx \lesssim \int_{\mathbb{R}^3} \langle x \rangle^{-k} |f|^2 dx, \quad (1.3)$$

where $\mathcal{R}$ is a Calderon–Zygmund operator. Unfortunately, when $k > 3$, $\langle x \rangle^{-k}$ does not belong to any $\mathcal{A}_p$ class (see [10]), which causes extra technical difficulties in estimating (1.3). To overcome it, we exploit the inherent divergence structure in nonlinear terms and use elementary integration by parts techniques, see Lemma 4.1.

1.4 Structure of the paper

The rest of the paper is organized as follows. In section 2, we will define the L^∞ mild solution of (1.1), and describe some basic properties. In section 3, we will establish a weighted-type estimate, which is the key tool in our proof. In section 4, we will give a rigorous proof of the main theorem 1.1. *Notations.* In this paper, we denote $A \lesssim B$ to mean $A \leq cB$ for some universal constant $c > 0$. We denote $A \approx B$ to mean that $A \lesssim B$ and $B \lesssim A$. We also use $A \ll B$ to denote that $A \leq \frac{1}{M}B$ for some large universal constant $M > 0$. We use the Japanese bracket $\langle \cdot \rangle$ to denote $\sqrt{1 + |\cdot|^2}$.

2 Preliminary

In this section, we will give a brief definition of L^∞ mild solutions of (1.1).

Definition 2.1. $u(t)$ is defined to be an L^∞ mild solution of the Cauchy problem (1.1), if $u \in L_{t,x}^\infty$ satisfies (1.1) in the distributional sense and the following integral equation

$$u(t) = e^{(t-s)\Delta} u(s) + \int_s^t e^{(t-\tau)\Delta} \mathbb{P}[\nabla \cdot (u \otimes u)(\cdot, \tau) + f(\cdot, \tau)] d\tau, \quad 0 \leq s \leq t \leq T. \quad (2.1)$$

Remark 2.1. Here $e^{t\Delta}$ is the standard heat semigroup, and $\mathbb{P}$ denotes the Leray projection on $\mathbb{R}^3$. Clearly, $\mathbb{P}$ is a Calderon–Zygmund type singular integral operator and can be formally written by

$$\mathbb{P} = I + \nabla(-\Delta)^{-1}(\nabla \cdot).$$

Remark 2.2. From the definition of mild solution (2.1), the pressure term can be recovered by

$$\nabla p = \nabla(-\Delta)^{-1} \partial_t \partial_j u^i u^j - \nabla(-\Delta)^{-1}(\nabla \cdot f),$$

therefore we are able to estimate the pressure in terms of the velocity field.

Remark 2.3. It is clear that (for instance, see [14]) (2.1) can be written in a convolution form. Explicitly,

$$\int_0^t e^{(t-s)\Delta} \mathbb{P}[(u \cdot \nabla)u](\cdot, s) ds = \int_0^t \int_{\mathbb{R}^3} (\nabla O(x-y, t-s) :: u \otimes u(y, s)) dy ds, \quad (2.2)$$

$$\int_0^t e^{(t-s)\Delta} \mathbb{P}(f)(\cdot, s) ds = \int_0^t \int_{\mathbb{R}^3} O(x-y, t-s) \cdot f(y, s) dy ds, \quad (2.3)$$

where the kernel satisfies following estimates:

$$|\nabla^k O(x, t)| \lesssim (|x| + \sqrt{t})^{-3-|k|}, \quad (2.4)$$

which implies (2.1) is well defined for $u \in L_{t,x}^\infty$.

3 Weighted L^2 Estimates

In this section, we will prove the following weighted inequality.

Theorem 3.1. Assume $u(x, t) \in C_c^\infty(\mathbb{R}^n \times (T_-, T^+))$, where $[T_-, T^+] \subset (0, \infty)$, and $T^+ \leq \tau$ for some universal small constant τ . Then for any large number $a > 0$ and any $k \geq 0$, the following inequality holds:

$$\begin{aligned} & \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)} \langle x \rangle^{-2k} \left((a+1)u^2 + |\nabla u|^2 \right) dx dt \\ & \lesssim \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a} \langle x \rangle^{-2k} |\partial_t u + \Delta u|^2 dx dt, \end{aligned} \quad (3.1)$$

where $h(t) = te^{-t}$.

Remark 3.1. The above inequality is a type of (time-space) weighted L^2 estimate. The corresponding weight in Theorem 3.1 satisfies the following facts:

- (i) The temporal weight $h(t)$ inspired from [8] is sufficiently large when time is small enough.
- (ii) For $k > \frac{3}{2}$, the spatial weight is integrable on the whole space. These two facts are crucial in our proof.

To prove Theorem 3.1, we need the following lemma.

Lemma 3.2. Assume A is a densely defined operator on a real Hilbert space. Then for each $x \in \mathcal{D}(A^2)$, the following holds:

$$\|Ax\|^2 \geq \langle [J, K]x, x \rangle, \quad (3.2)$$

where $J = \frac{A+A^t}{2}$ is the symmetric part of A , and $K = \frac{A-A^t}{2}$ is the skew symmetric part of A , $[J, K] = JK - KJ$.

Proof. By direct computation,

$$\begin{aligned}\|Ax\|^2 &= \|(J+K)x\|^2 = (J+K)x, (J+K)x \\ &= \|Jx\|^2 + \|Kx\|^2 + \langle Jx, Kx \rangle + \langle Kx, Jx \rangle \\ &\geq \langle Jx, Kx \rangle + \langle Kx, Jx \rangle \\ &= \langle [J, K]x, x \rangle.\end{aligned}$$

■

Now we are ready to prove Theorem 3.1.

Proof of Theorem 3.1. We consider

$$\begin{aligned}L &:= h(t)^{-(a+1)}(1+|x|^2)^{-k} \cdot (\partial_t + \Delta) \cdot h(t)^{a+1}(1+|x|^2)^k \\ &= \partial_t + \Delta + (a+1) \frac{h'(t)}{h(t)} + \frac{4kx}{1+|x|^2} \cdot \nabla + \frac{2kn}{1+|x|^2} + \frac{4k(k-1)|x|^2}{(1+|x|^2)^2}.\end{aligned}$$

Let $A = tL$. Then we can decompose A as:

$$\begin{aligned}A &= t\partial_t + t\Delta + (a+1)(1-t) + \frac{4ktx}{1+|x|^2} \cdot \nabla + \frac{2knt}{1+|x|^2} + \frac{4k(k-1)t|x|^2}{(1+|x|^2)^2} \\ &= \left(t\Delta + \frac{4k^2t|x|^2}{(1+|x|^2)^2} + (a+1)(1-t) - \frac{1}{2} \right) \\ &\quad + \left(t\partial_t + \frac{4ktx}{1+|x|^2} \cdot \nabla + 2kt \left(\frac{n}{1+|x|^2} - \frac{2|x|^2}{(1+|x|^2)^2} \right) + \frac{1}{2} \right) \\ &=: J + K.\end{aligned}$$

It is direct to see that J is symmetric since Δ is a symmetric operator. Besides, using integration by parts, we can show that $t\partial_t + \frac{1}{2}$ and $\frac{4ktx}{1+|x|^2} \cdot \nabla + 2kt \left(\frac{n}{1+|x|^2} - \frac{2|x|^2}{(1+|x|^2)^2} \right)$ are skew-symmetric operators, hence K is skew-symmetric. Furthermore, the commutator of J, K can be computed as follows:

$$\begin{aligned}[J, K] &= (a+1)t - t\Delta - \frac{4k^2t|x|^2}{(1+|x|^2)^2} \\ &\quad + \frac{8kt^2}{1+|x|^2} \Delta - \frac{16kt^2}{(1+|x|^2)^2} [(x \otimes x) :: \nabla^2] \\ &\quad + t^2 V_1(x) x \cdot \nabla \\ &\quad + t^2 V_2(x),\end{aligned}$$

where

$$\begin{aligned}V_1(x) &= \frac{64k|x|^2}{(1+|x|^2)^3} - \frac{16kn+32k}{(1+|x|^2)^2}, \\ V_2(x) &= -\frac{(4n+8)kn}{(1+|x|^2)^2} + \frac{(32kn+64k-32k^3)|x|^2}{(1+|x|^2)^3} + \frac{(64k^3-96k)|x|^4}{(1+|x|^2)^4}.\end{aligned}$$

Let $v(x, t) = h(t)^{-(a+1)}(1 + |x|^2)^{-k}u(x, t)$, and applying lemma 3.2, we obtain that

$$\begin{aligned} \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a}(1 + |x|^2)^{-2k} |\partial_t u + \Delta u|^2 dx dt &\approx \|Av\|_{L^2}^2 \\ &\geq \langle [J, K]v, v \rangle \\ &=: \sum_{j=1}^6 I_j, \end{aligned}$$

where

$$\begin{aligned} I_1 &= \int_{\mathbb{R}^n \times (T_-, T^+)} (a+1)tv^2 dx dt, \\ I_2 &= \int_{\mathbb{R}^n \times (T_-, T^+)} -tv\Delta v dx dt, \\ I_3 &= \int_{\mathbb{R}^n \times (T_-, T^+)} -\frac{4k^2 t|x|^2}{(1 + |x|^2)^2} v^2 dx dt, \\ I_4 &= \int_{\mathbb{R}^n \times (T_-, T^+)} \frac{8kt^2}{1 + |x|^2} v \left(\Delta v - \left[\frac{2(x \otimes x)}{1 + |x|^2} :: \nabla^2 \right] v \right) dx dt, \\ I_5 &= \int_{\mathbb{R}^n \times (T_-, T^+)} t^2 V_1(x) v(x \cdot \nabla) v dx dt, \\ I_6 &= \int_{\mathbb{R}^n \times (T_-, T^+)} t^2 V_2(x) v^2 dx dt. \end{aligned}$$

It is easy to check that

$$\begin{aligned} I_1 &\gtrsim (a+1) \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)}(1 + |x|^2)^{-2k} |u|^2 dx dt; \\ I_3 &\lesssim - \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)}(1 + |x|^2)^{-2k} |u|^2 dx dt. \end{aligned}$$

For I_2 , integrating by parts, we have

$$\begin{aligned} I_2 &\gtrsim \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)} \left| \nabla \left((1 + |x|^2)^{-k} u \right) \right|^2 dx dt \\ &\gtrsim \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)}(1 + |x|^2)^{-2k} |\nabla u|^2 dx dt \\ &\quad - 2 \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)}(1 + |x|^2)^{-2k} |u|^2 dx dt. \end{aligned}$$

For I_4 , integrating by parts again, we have

$$\begin{aligned} |I_4| &\lesssim \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a}(1 + |x|^2)^{-1} \left| \nabla \left((1 + |x|^2)^{-k} u \right) \right|^2 dx dt \\ &\lesssim \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a}(1 + |x|^2)^{-2k} |\nabla u|^2 dx dt + \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a}(1 + |x|^2)^{-2k} |u|^2 dx dt. \end{aligned}$$

Similarly, integrating by parts, and applying Cauchy–Schwarz inequality, we can also obtain

$$|I_5 + I_6| \lesssim \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a}(1 + |x|^2)^{-2k} |\nabla u|^2 dx dt + \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a}(1 + |x|^2)^{-2k} |u|^2 dx dt.$$

Combining the above all estimates, we have

$$\begin{aligned}
& \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a} (1 + |x|^2)^{-2k} |\partial_t u + \Delta u|^2 dx dt \\
& \gtrsim (a+1) \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)} (1 + |x|^2)^{-2k} |u|^2 dx dt \\
& \quad + \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)} (1 + |x|^2)^{-2k} |\nabla u|^2 dx dt \\
& \quad - C \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)} (1 + |x|^2)^{-2k} |u|^2 dx dt \\
& \quad - C \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a} (1 + |x|^2)^{-2k} |\nabla u|^2 dx dt,
\end{aligned}$$

where $C > 0$ is some universal constant. Using the fact that a is large enough and T^+ is small enough so that $h(T^+)^{-1} \approx (T^+)^{-1} \gg C$, we have

$$\begin{aligned}
& \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a} (1 + |x|^2)^{-2k} |\partial_t u + \Delta u|^2 dx dt \\
& \gtrsim (a+1) \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)} (1 + |x|^2)^{-2k} |u|^2 dx dt \\
& \quad + \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)} (1 + |x|^2)^{-2k} |\nabla u|^2 dx dt.
\end{aligned}$$

By approximating with compactly supported smooth functions, it holds that

Corollary 3.3. Assume $u \in H_0^1((T_-, T^+); W^{2,p}(\mathbb{R}^3))$, where $2 \leq p \leq \infty$, then (3.1) still holds for $k > 3\left(\frac{1}{2} - \frac{1}{p}\right)$.

Proof. Let η be a smooth spatial cut-off function such that $\eta \equiv 1$ for $|x| \leq 1$, and $\eta \equiv 0$ for $|x| \geq 2$. Let $\eta_m(x) = \eta(\frac{x}{m})$. Replacing u by $\eta_m u$, and applying Theorem 3.1 we obtain

$$\begin{aligned}
& \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)} \langle x \rangle^{-2k} \left((a+1)u^2 + |\nabla u|^2 \right) dx dt \\
& = \lim_{m \rightarrow \infty} \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-(2a+1)} \langle x \rangle^{-2k} \left((a+1)(\eta_m u)^2 + |\nabla(\eta_m u)|^2 \right) dx dt \\
& \lesssim \lim_{m \rightarrow \infty} \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a} \langle x \rangle^{-2k} |\partial_t(\eta_m u) + \Delta(\eta_m u)|^2 dx dt \\
& \lesssim \lim_{m \rightarrow \infty} \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a} \langle x \rangle^{-2k} |\eta_m|^2 |\partial_t u + \Delta u|^2 dx dt \\
& \quad + \lim_{m \rightarrow \infty} \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a} \langle x \rangle^{-2k} (|\Delta \eta_m|^2 |u|^2 + |\nabla \eta_m \nabla u|^2) dx dt \\
& \lesssim \int_{\mathbb{R}^n \times (T_-, T^+)} h(t)^{-2a} \langle x \rangle^{-2k} |\partial_t u + \Delta u|^2 dx dt.
\end{aligned}$$

4 Proof of the Main Result

In this section, we will prove the main theorem of this paper.

Proof of Theorem 1.1. Assume u_1, u_2 satisfy the assumption of Theorem 1.1. We choose T^+ sufficiently small so that T^+ satisfies the condition of Theorem 3.1 and in addition

$$T^+ \ll \left[\sum_{i=1,2} (\|u_i\|_{L^\infty_{t,x}}^2 + \|\nabla \times u_i\|_{L^\infty_{t,x}}^2) \right]^{-1}, \quad (4.1)$$

we also choose $T_- = \frac{T^+}{2}$. Let $u(x, t) = (u_1 - u_2)(x, T + T_- - t)$, then u satisfies the following equation:

$$\begin{cases} \partial_t u + \Delta u + \mathbb{P} \nabla \cdot (u_1 \otimes u + u \otimes u_2) = 0, & (x, t) \in \mathbb{R}^3 \times [T_-, T + T_-], \\ \nabla \cdot u = 0, \\ u(\cdot, T_-) = 0. \end{cases} \quad (4.2)$$

Then by Section 3 in [14], u is sufficiently smooth. Choose a temporal cut-off function $\chi \in C^\infty[T_-, T^+]$ such that $\chi(t) = 1$, when $t \in [T_-, T_- + \delta]$, and $\chi(t) \equiv 0$ when $t \in [T_- + 2\delta, T^+]$, where $\delta = \frac{T^+}{8}$. Applying Corollary 3.3 on $\chi(t)u(x, t)$ with $\frac{3}{2} < k < \frac{5}{2}$, we obtain

$$\begin{aligned} & \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-(2a+1)} \langle x \rangle^{-2k} ((a+1)|u|^2 + |\nabla u|^2) dx dt \\ & \lesssim \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |\partial_t u + \Delta u|^2 dx dt \\ & \quad + \int_{(T_-, T^+) \times \mathbb{R}^3} |\chi'(t)|^2 h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 dx dt \\ & \lesssim \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} \left| \mathbb{P} \nabla \cdot (u_1 \otimes u + u \otimes u_2) \right|^2 dx dt \\ & \quad + \int_{(T_-, T^+) \times \mathbb{R}^3} |\chi'(t)|^2 h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 dx dt \\ & := A + B. \end{aligned} \quad (4.3)$$

Firstly, we estimate A . It turns out that A can be absorbed in the left-hand side. Indeed,

$$\begin{aligned} A & \lesssim \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u_1 \cdot \nabla u + u \cdot \nabla u_2|^2 dx dt \\ & \quad + \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |\mathcal{R} \nabla \cdot (u_1 \otimes u)|^2 dx dt \\ & \quad + \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |\mathcal{R} \nabla \cdot (u \otimes u_2)|^2 dx dt \\ & := A_1 + A_2 + A_3, \end{aligned} \quad (4.4)$$

where $\mathcal{R} = \nabla(-\Delta)^{-1}\nabla \cdot$ is the zeroth-order singular integral operator. Note that A_1 can be estimated directly,

$$\begin{aligned} A_1 & \lesssim \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u_1|^2 |\nabla u|^2 dx dt \\ & \quad + \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 |\nabla u_2|^2 dx dt := A_{11} + A_{12}. \end{aligned}$$

Clearly,

$$A_{11} \lesssim \|u_1\|_{L^\infty_{t,x}}^2 \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |\nabla u|^2 dx dt.$$

To ensure ∇u_2 is not involved in the estimate of A_{12} , we have to rewrite the term $|\nabla u_2|^2$. Indeed,

$$\begin{aligned} A_{12} &= \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 |\nabla u_2|^2 dx dt \\ &= \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 (\Delta(|u_2|^2) - u_2 \cdot \Delta u_2) dx dt := A_{13} + A_{14}. \end{aligned}$$

Firstly, we estimate the term A_{14} . Note that $-\Delta u_2 = \nabla \times \nabla \times u_2$ since $\nabla \cdot u_2 = 0$, integrating by parts, we have

$$\begin{aligned} A_{14} &= \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 u_2 \cdot (\nabla \times \nabla \times u_2) dx dt \\ &= - \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \nabla \times (\langle x \rangle^{-2k} |u|^2 u_2) \cdot \nabla \times u_2 dx dt \\ &\lesssim \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-(2k+1)} |u|^2 |u_2| \cdot |\nabla \times u_2| dx dt \\ &\quad + \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u| \cdot |\nabla u| \cdot |u_2| \cdot |\nabla \times u_2| dx dt \\ &\quad + \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 |\nabla \times u_2|^2 dx dt \\ &\lesssim (\|u_2\|_{L_{t,x}^\infty}^2 + \|\nabla \times u_2\|_{L_{t,x}^\infty}^2) \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} (|u|^2 + |\nabla u|^2) dx dt. \end{aligned}$$

For A_{13} , integrating by parts again we have

$$\begin{aligned} A_{13} &= \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 (\Delta(|u_2|^2)) dx dt \\ &\leq \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \left| \nabla (\langle x \rangle^{-2k} |u|^2) \right| \cdot \left| \nabla(|u_2|^2) \right| dx dt \\ &\lesssim \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-(2k+2)} |u|^2 |u_2| \cdot |\nabla u_2| dx dt \\ &\quad + \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u| \cdot |\nabla u| \cdot |u_2| \cdot |\nabla u_2| dx dt \\ &\lesssim \epsilon \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 |\nabla u_2|^2 dx dt \\ &\quad + C_\epsilon \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} |u_2|^2 (|u|^2 + |\nabla u|^2) dx dt \\ &\lesssim \epsilon A_{12} + C_\epsilon \|u_2\|_{L_{t,x}^\infty}^2 \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} (|u|^2 + |\nabla u|^2) dx dt, \end{aligned}$$

where ϵ is sufficiently small and C_ϵ is a constant only depends on ϵ . Combining the above computations, we have

$$A_{12} \lesssim (\|u_2\|_{L_{t,x}^\infty}^2 + \|\nabla \times u_2\|_{L_{t,x}^\infty}^2) \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} (|u|^2 + |\nabla u|^2) dx dt.$$

To estimate A_2 and A_3 , we prove the following lemma: ■

Lemma 4.1. for $0 \leq k < \frac{5}{2}$, the following inequality holds:

$$\int_{\mathbb{R}^3} \langle x \rangle^{-2k} |\mathcal{R} \nabla \cdot f|^2 dx \lesssim \int_{\mathbb{R}^3} \langle x \rangle^{-2k} (|\nabla f|^2 + |f|^2) dx. \quad (4.5)$$

Remark 4.1. Note that for $k \geq \frac{3}{2}$, $(1 + |x|^2)^{-k}$ does not belong to any $\mathcal{A}_p$ class, however, the divergence structure enables us to improve the inequality to $k < \frac{5}{2}$, which makes the RHS of (4.5) integrable for f with no spatial decay.

By the definition of $\mathcal{R}$ and divergence, for $1 \leq i, j, l \leq 3$, we define

$$\mathcal{R}_{ijl} = \mathcal{F}^{-1} \frac{\xi_i \xi_j}{|\xi|^2} \mathcal{F} \partial_{x_l},$$

where $\mathcal{F}$ denotes the standard Fourier transform. Then with a slight abuse of notations, we shall only estimate

$$I_{ijl} = \int_{\mathbb{R}^3} (1 + |x|^2)^{-k} |\mathcal{R}_{ijl} f|^2 dx.$$

By the representation formula of homogeneous singular integral operator, for each $1 \leq i, j \leq 3$, there exists a smooth function Ω on $\mathbb{S}^2$ with mean zero such that

$$\begin{aligned} \mathcal{R}_{ijl} f &= \frac{\Omega(y/|y|)}{|y|^3} * \partial_{x_l} f := \lim_{\lambda \rightarrow 0^+} \int_{|y| \geq \lambda} \frac{\Omega(y/|y|)}{|y|^3} \partial_{x_l} f(\cdot - y) dy \\ &= \int_{\mathbb{R}^3} \frac{\Omega(y/|y|)}{|y|^3} (\partial_{x_l} f(\cdot - y) - \partial_{x_l} f(\cdot)) dy. \end{aligned}$$

Choosing $\varphi \in C_c^\infty(B_2)$ and $\varphi \equiv 1$ in B_1 , and integrating by parts, we can decompose $\mathcal{R}_{ijl}$ into

$$\begin{aligned} \mathcal{R}_{ijl} f &= \left[\frac{\Omega(y/|y|)}{|y|^3} \varphi(y) \right] * \partial_{x_l} f - \partial_{y_l} \left[\frac{\Omega(y/|y|)}{|y|^3} (1 - \varphi(y)) \right] * f \\ &:= \mathcal{R}_{ijl1} f + \mathcal{R}_{ijl2} f. \end{aligned}$$

The virtue of this decomposition is that $\mathcal{R}_{ijl1}$ is a localized operator while the kernel of $\mathcal{R}_{ijl2}$ is integrable. Hence we compute

$$\begin{aligned} I_{ijl} &= \int_{\mathbb{R}^3} (1 + |x|^2)^{-k} |\mathcal{R}_{ijl1} f + \mathcal{R}_{ijl2} f|^2 dx \\ &\lesssim \int_{\mathbb{R}^3} (1 + |x|^2)^{-k} \left| \mathcal{R}_{ijl1} f \right|^2 dx \\ &\quad + \int_{\mathbb{R}^3} (1 + |x|^2)^{-k} \left| \mathcal{R}_{ijl2} f \right|^2 dx \\ &:= I_{ijl1} + I_{ijl2}. \end{aligned}$$

We will estimate I_{ijl1} , I_{ijl2} separately. Estimates of I_{ijl1} . First, note that

$$\begin{aligned} (1 + |x|^2)^{-k/2} &= (1 + |x - y|^2)^{-k/2} - \int_0^1 \frac{d}{dt} (1 + |x - ty|^2)^{-k/2} dt \\ &= (1 + |x - y|^2)^{-k/2} - \int_0^1 (1 + |x - ty|^2)^{-(k+2)/2} (x - ty) \cdot y dt. \end{aligned}$$

We decompose

$$\begin{aligned} I_{ijl1} &\lesssim \int_{\mathbb{R}^3} \left| \int_{\mathbb{R}^3} (1+|x-y|^2)^{-k/2} \frac{\Omega(y/|y|)}{|y|^3} \varphi(y) \nabla f(x-y) dy \right|^2 dx \\ &\quad + \int_{\mathbb{R}^3} \left| \int_{\mathbb{R}^3} \int_0^1 (1+|x-ty|^2)^{-(k+1)/2} dt \frac{\Omega(y/|y|)}{|y|^2} \varphi(y) \nabla f(x-y) dy \right|^2 dx \\ &:= I_{11} + I_{12}. \end{aligned}$$

By the L^2 boundedness of the singular integral integral,

$$I_{11} \lesssim \int_{\mathbb{R}^3} (1+|x|^2)^{-k} |\nabla f|^2 dx.$$

For I_{12} , since for $|y| \leq 2$, we have

$$1+|x-y| \lesssim 1+|x-ty|,$$

by Minkowski's inequality, we have

$$\begin{aligned} I_{12} &\lesssim \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \left| (1+|x-y|^2)^{-(k+1)/2} \partial_{y_k} f(x-y) \right|^2 \frac{\varphi(y)}{|y|^2} dy dx \\ &\lesssim \int_{\mathbb{R}^3} (1+|x|^2)^{-k} |\nabla f|^2 dx. \end{aligned}$$

Estimates of I_{ijl2} . Note that

$$\partial_{y_k} \left[\frac{\Omega(y/|y|)}{|y|^3} (1-\varphi(y)) \right] \lesssim (1+|y|^2)^{-2},$$

we have

$$\begin{aligned} I_{ijl2} &\lesssim \int_{\mathbb{R}^3} (1+|x|^2)^{-k} \left| \int_{\mathbb{R}^3} f(x-y) (1+|y|^2)^{-2} dy \right|^2 dx \\ &= \int_{\mathbb{R}^3} \left| \int_{\mathbb{R}^3} f(x-y) (1+|x|^2)^{-\frac{k}{2}} (1+|y|^2)^{-2} dy \right|^2 dx \\ &\lesssim \int_{\mathbb{R}^3} \left| \int_{\mathbb{R}^3} f(x-y) (1+|x-y|^2)^{-\frac{k}{2}} \left((1+|y|^2)^{-2} + (1+|y|^2)^{-(2-\frac{k}{2})} (1+|x|^2)^{-\frac{k}{2}} \right) dy \right|^2 dx \\ &\lesssim \int_{\mathbb{R}^3} \left| \int_{\mathbb{R}^3} f(x-y) (1+|x-y|^2)^{-\frac{k}{2}} (1+|y|^2)^{-2} dy \right|^2 dx \\ &\quad + \int_{\mathbb{R}^3} \left| \int_{\mathbb{R}^3} f(x-y) (1+|x-y|^2)^{-\frac{k}{2}} (1+|y|^2)^{-(2-\frac{k}{2})} (1+|x|^2)^{-\frac{k}{2}} dy \right|^2 dx \\ &:= I_{21} + I_{22}, \end{aligned}$$

where in the second inequality we used the fact that

$$(1+|x|^2)^{-\frac{k}{2}} (1+|y|^2)^{-2} \lesssim (1+|x-y|^2)^{-\frac{k}{2}} \left((1+|y|^2)^{-2} + (1+|y|^2)^{-(2-\frac{k}{2})} (1+|x|^2)^{-\frac{k}{2}} \right).$$

For I_{21} , using Young's inequality we have

$$I_{21} \lesssim \int_{\mathbb{R}^3} (1+|x|^2)^{-k} f^2(x) dx.$$

For I_{22} , since $k < \frac{5}{2}$, using Cauchy–Schwarz inequality we have

$$\begin{aligned} I_{22} &= \int_{\mathbb{R}^3} (1 + |x|^2)^{-k} \left(\int_{\mathbb{R}^3} f(x-y) (1 + |x-y|^2)^{-\frac{k}{2}} (1 + |y|^2)^{-(2-\frac{k}{2})} dy \right)^2 dx \\ &\lesssim \left(\int_{\mathbb{R}^3} (1 + |x|^2)^{-\frac{5}{2}} dx \right)^{\frac{2k}{5}} \left\| \int_{\mathbb{R}^3} f(x-y) (1 + |x-y|^2)^{-\frac{k}{2}} (1 + |y|^2)^{-(2-\frac{k}{2})} dy \right\|_{L_x^{\frac{10}{5-2k}}(\mathbb{R}^3)}^2. \end{aligned}$$

Then using Young's inequality, since $k < \frac{5}{2}$ implies $\frac{5(4-k)}{5-k} > 3$, we have

$$\begin{aligned} I_{22} &\lesssim \left\| \int_{\mathbb{R}^3} f(x-y) (1 + |x-y|^2)^{-\frac{k}{2}} (1 + |y|^2)^{-(2-\frac{k}{2})} dy \right\|_{L_x^{\frac{10}{5-2k}}(\mathbb{R}^3)}^2 \\ &\lesssim \int_{\mathbb{R}^3} (1 + |x|^2)^{-k} f^2(x) dx \left\| (1 + |y|^2)^{-(2-\frac{k}{2})} \right\|_{L_x^{\frac{5}{5-k}}(\mathbb{R}^3)}^2 \\ &\lesssim \int_{\mathbb{R}^3} (1 + |x|^2)^{-k} f^2(x) dx. \end{aligned}$$

Combining the above estimates the lemma 4.1 is proved. The estimates of A_2 and A_3 now follows from choosing $f = u_1 \otimes u$ and $f = u \otimes u_2$. We have

$$A_2 \lesssim \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} (|u_1|^2 |u|^2 + |u_1|^2 |\nabla u|^2 + |\nabla u_1|^2 |u|^2) dx dt.$$

Similarly,

$$A_3 \lesssim \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-2a} \langle x \rangle^{-2k} (|u_2|^2 |u|^2 + |u_2|^2 |\nabla u|^2 + |\nabla u_2|^2 |u|^2) dx dt.$$

Using the same techniques as in estimating A_{12} , we also have

$$\begin{aligned} A_2 + A_3 &\lesssim \sum_{i=1,2} (\|u_i\|_{L_{t,x}^\infty}^2 + \|\nabla \times u_i\|_{L_{t,x}^\infty}^2) \int_{(T_-, T^+) \times \mathbb{R}^3} \chi(t)^2 h(t)^{-2a} \langle x \rangle^{-2k} (|u|^2 + |\nabla u|^2) dx dt. \end{aligned}$$

All in all, we have

$$\begin{aligned} A &= A_1 + A_2 + A_3 \\ &\lesssim \sum_{i=1,2} (\|u_i\|_{L_{t,x}^\infty}^2 + \|\nabla \times u_i\|_{L_{t,x}^\infty}^2) \int_{(T_-, T^+) \times \mathbb{R}^3} \chi(t)^2 h(t)^{-2a} \langle x \rangle^{-2k} (|u|^2 + |\nabla u|^2) dx dt \\ &\lesssim T^+ \sum_{i=1,2} (\|u_i\|_{L_{t,x}^\infty}^2 + \|\nabla \times u_i\|_{L_{t,x}^\infty}^2) \int_{(T_-, T^+) \times \mathbb{R}^3} \chi(t)^2 h(t)^{-(2a+1)} \langle x \rangle^{-2k} (|u|^2 + |\nabla u|^2) dx dt. \end{aligned}$$

By (4.1) we have

$$A \ll \int_{(T_-, T^+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-(2a+1)} \langle x \rangle^{-2k} (|u|^2 + |\nabla u|^2) dx dt.$$

As a result, A can be absorbed by the left-hand side of (4.3) and we have

$$\begin{aligned} & \int_{(T_-, T_+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-(2a+1)} \langle x \rangle^{-2k} ((a+1)|u|^2 + |\nabla u|^2) dx dt \\ & \lesssim \int_{(T_-, T_+) \times \mathbb{R}^3} |\chi'(t)|^2 h(t)^{-2a} \langle x \rangle^{-2k} |u|^2 dx dt. \end{aligned}$$

By the definition of χ , we have

$$\begin{aligned} & \left(h(T_- + \delta/2) \right)^{-(2a+1)} \int_{(T_-, T_- + \frac{\delta}{2}) \times \mathbb{R}^3} \langle x \rangle^{-2k} |u|^2 dx dt \\ & \leq \int_{(T_-, T_+) \times \mathbb{R}^3} \chi^2(t) h(t)^{-(2a+1)} \langle x \rangle^{-2k} |u|^2 dx dt \\ & \lesssim \int_{(T_-, T_+) \times \mathbb{R}^3} |\chi'(t)|^2 h(t)^{-(2a+1)} \langle x \rangle^{-2k} |u|^2 dx dt \\ & \lesssim \left(h(T_- + \delta) \right)^{-(2a+1)} \int_{(T_- + \delta, T_- + 2\delta) \times \mathbb{R}^3} \langle x \rangle^{-2k} |u|^2 dx dt, \end{aligned}$$

or equivalently,

$$\int_{(T_-, T_- + \frac{\delta}{2}) \times \mathbb{R}^3} \langle x \rangle^{-2k} |u|^2 dx dt \lesssim \left(\frac{h(T_- + \delta/2)}{h(T_- + \delta)} \right)^{2a+1} \int_{(T_- + \delta, T_- + 2\delta) \times \mathbb{R}^3} \langle x \rangle^{-2k} |u|^2 dx dt.$$

Letting $a \rightarrow \infty$ and using the fact that h is a strictly increasing function in $[0, \frac{1}{2}]$, we obtain $u \equiv 0$ on $[T_-, T_- + \frac{\delta}{2}]$, or equivalently, $u_1 \equiv u_2$ on $[T - \frac{\delta}{2}, T]$. Clearly, δ is independent of the terminal time T . Repeating the above procedure for finite times, we obtain u_1 must coincide with u_2 at any time $t < T$, which finishes the proof of Theorem 1.1.

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